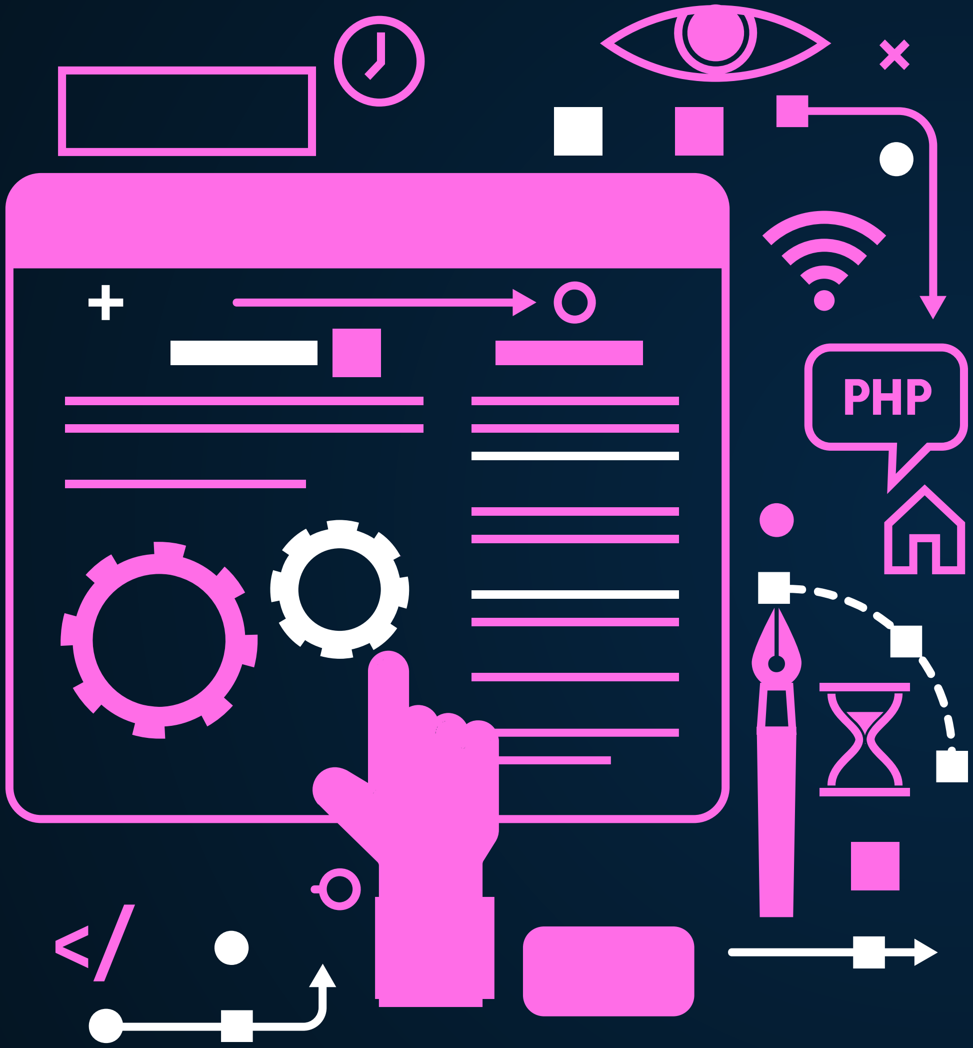

The Gender Wage Gap: Model, Analysis, and Solutions

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TABLE OF CONTENTS

The Global Issue

Cause

- Occupational segregation
- Discrimination and bias

- Independence
- Decreased job satisfaction and motivation

Effect

Significance

- Financial well-being
- Economic growth and development

01

Objective

To what extent do the different factors affect salary of male and female?

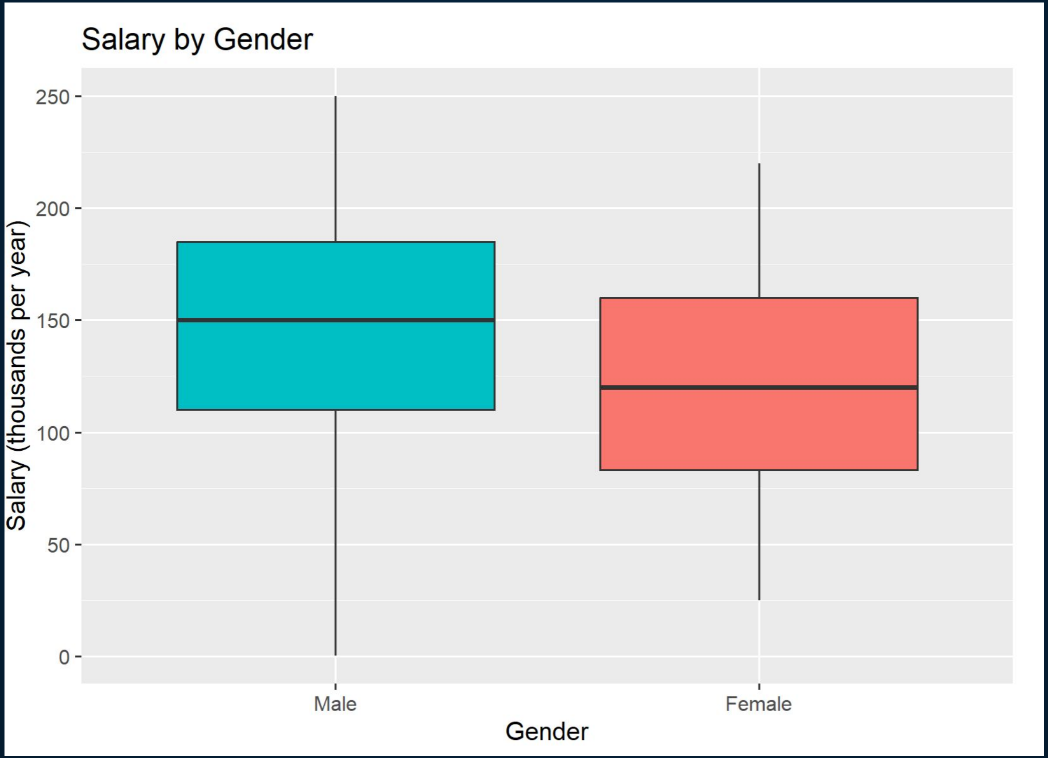
How might we address the global issue of the gender wage gap?

02

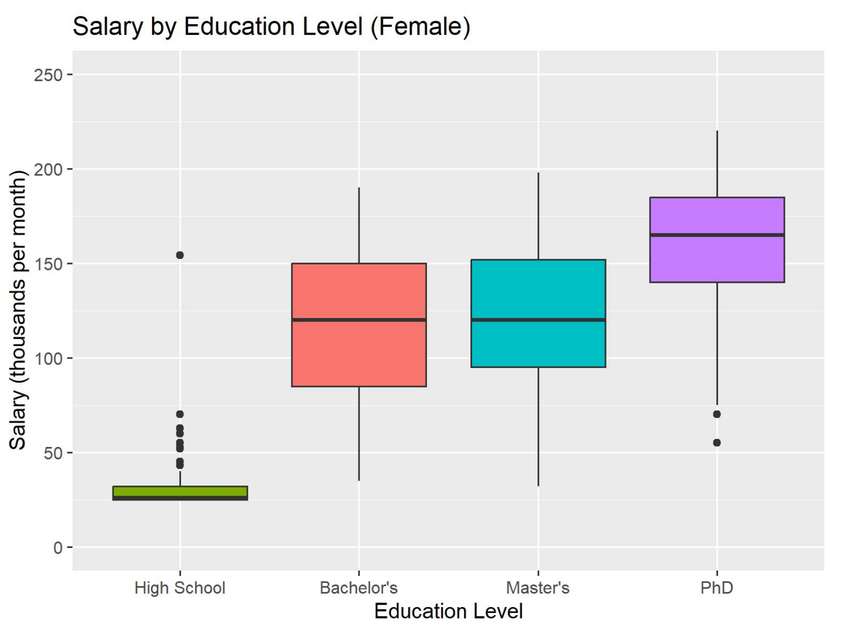
Significance (EDA)

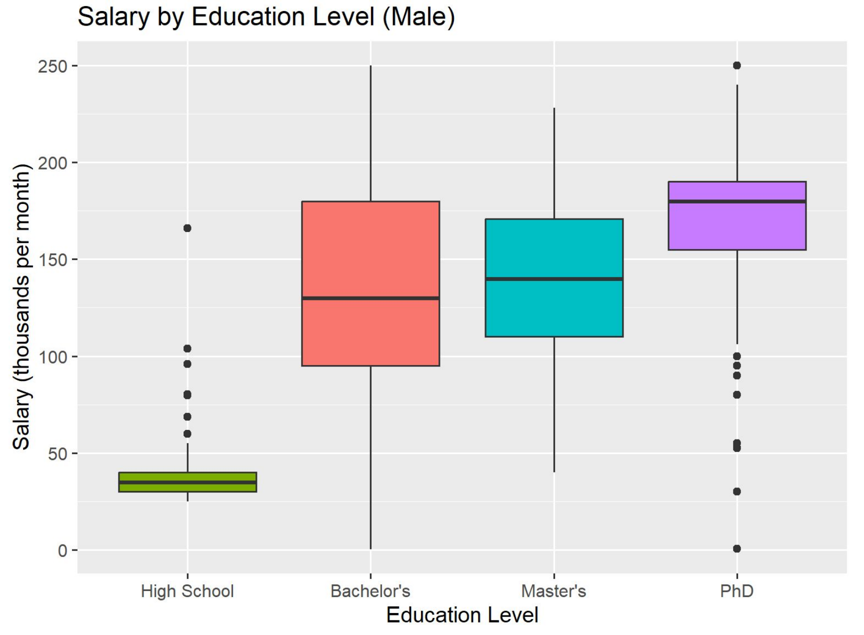
Does the gender wage gap exist?

Box Plot

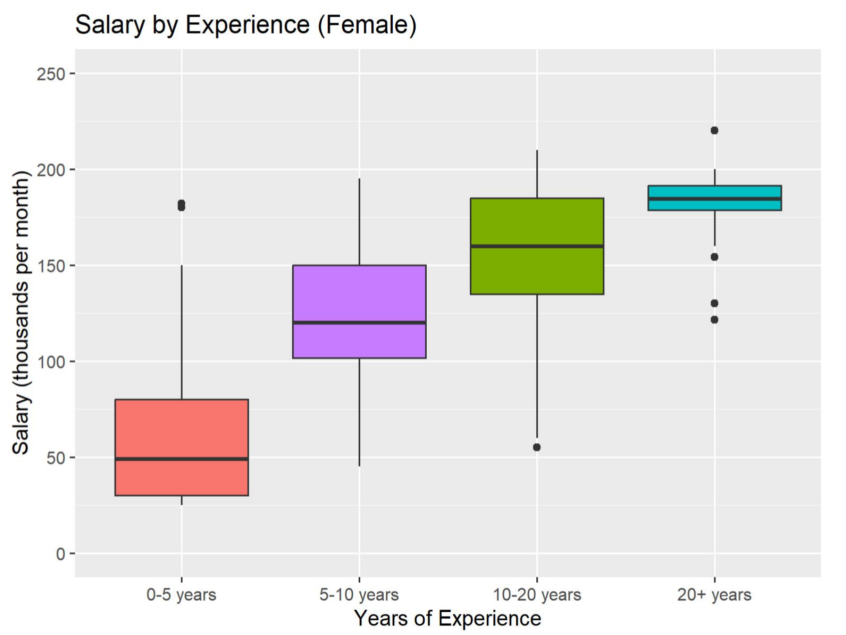

Salary by Gender
Salary (thousands per year)
250
200
150
100
50
0
Male
Female
Gender

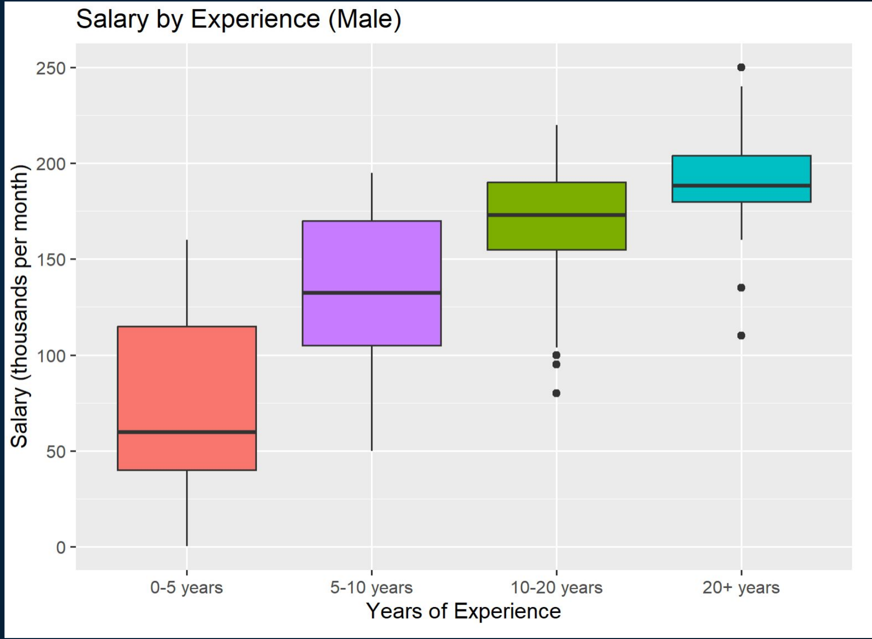
Box Plot (Salary vs Education)





Box Plot (Salary vs Experience)



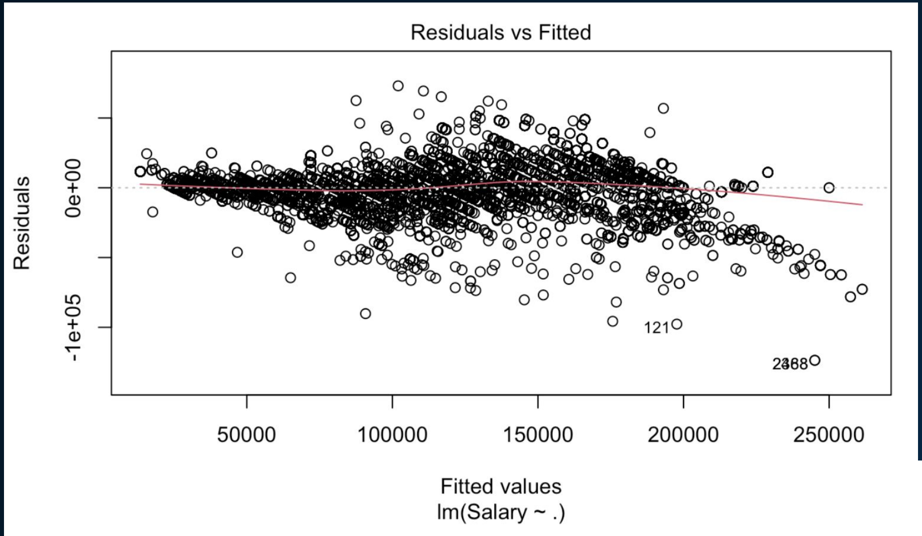


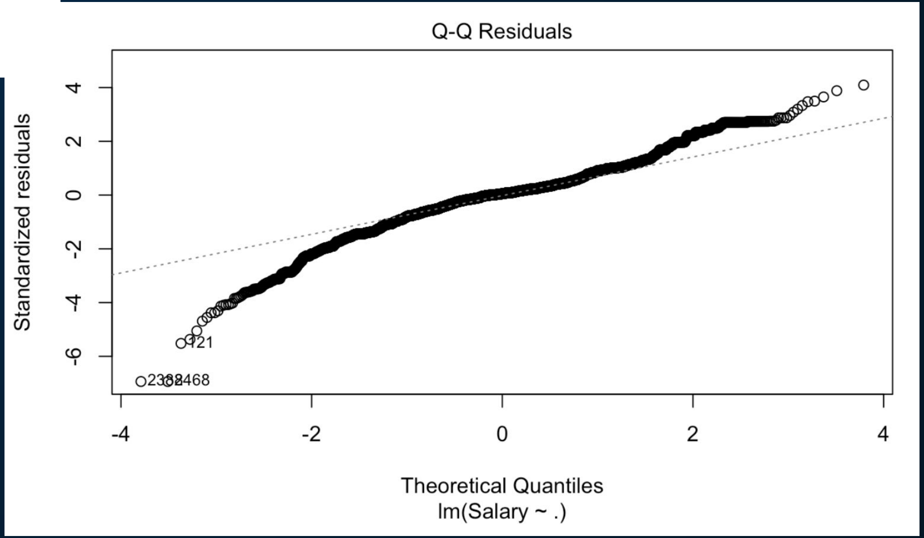
Multiple Regression

```
Response: Salary
                        Sum Sq   Df  F value     Pr(>F)    
Age                 3.4856e+10    1  108.723 < 2.2e-16 ***
Gender              8.0536e+09    2   12.560 3.594e-06 ***
Education.Level     9.2070e+11    6  478.641 < 2.2e-16 ***
Job.Title           2.1489e+12  190   35.279 < 2.2e-16 ***
Years.of.Experience 3.4013e+11    1 1060.950 < 2.2e-16 ***
Residuals           2.0829e+12 6497                       
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
GenderMale                      1531.0      512.5   2.987 0.002827 **
```

Reference Category: Female


Residuals vs Fitted
Residuals
0e+00
-1e+05
121
2388
50000
100000
150000
200000
250000
Fitted values
lm(Salary ~ .)


Q-Q Residuals
Standardized residuals
4
2
0
-2
-4
-6
121
2388
-4
-2
0
2
4
Theoretical Quantiles
lm(Salary ~ .)

03

Relaxed LASSO

What factors affect wage for males and females?

Dataset

Current Population Survey (CPS) of US National Data (1981~2013)

Contains Legal, Social, and Economic Factors (eg. age, education level, hours worked)

Data Cleaning

1) NA Values
2) Columns with Single Variable
3) Industries (Binary)
4) Repeated Variables

Men

```
Anova Table (Type II tests)

Response: incwage
              Sum Sq     Df    F value    Pr(>F)    
year      3.4734e+16      1 11077.8225 < 2.2e-16 ***
numprec   8.0022e+14      1   255.2181 < 2.2e-16 ***
statefip  2.9813e+15     42    22.6390 < 2.2e-16 ***
relate    1.1579e+14     11     3.3572 0.0001186 ***
age       5.1699e+15      1  1648.8661 < 2.2e-16 ***
race      3.9819e+15      3   423.3242 < 2.2e-16 ***
marst     5.7340e+14      5    36.5755 < 2.2e-16 ***
sch       5.3443e+16     19   897.1075 < 2.2e-16 ***
classwkr  3.7093e+15      6   197.1705 < 2.2e-16 ***
wkswork1  6.3438e+15      1  2023.2786 < 2.2e-16 ***
uhrswork  1.7063e+16      1  5441.9338 < 2.2e-16 ***
ftype     5.0152e+13      3     5.3317 0.0011369 ** 
ft        5.0834e+13      1    16.2127 5.664e-05 ***
Residuals 5.5122e+17 175805                         
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Women

```
Anova Table (Type II tests)

Response: incwage
              Sum Sq     Df   F value    Pr(>F)    
year      1.1534e+16      1 9749.0682 < 2.2e-16 ***
numprec   2.1779e+13      1   18.4085 1.784e-05 ***
statefip  1.5365e+15     42   30.9209 < 2.2e-16 ***
relate    5.2613e+13     11    4.0427 6.024e-06 ***
age       1.0386e+15      1  877.8564 < 2.2e-16 ***
race      6.3245e+14      3  178.1888 < 2.2e-16 ***
marst     1.4470e+14      5   24.4602 < 2.2e-16 ***
sch       2.0590e+16     19  915.9726 < 2.2e-16 ***
classwkr  1.7801e+15      6  250.7689 < 2.2e-16 ***
wkswork1  3.6059e+15      1 3047.8635 < 2.2e-16 ***
uhrswork  6.7155e+15      1 5676.1421 < 2.2e-16 ***
ftype     4.1429e+13      3   11.6724 1.210e-07 ***
ft        2.4855e+13      1   21.0082 4.577e-06 ***
Residuals 1.9908e+17 168266                        
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Accurac y

Output

Men

```
mean((predict(fit1, data.test1) - data.test1$incwage)^2)
#mean square error
```

MSE =
2120026701

Women

```
mean((predict(fit2, data.test2) - data.test2$incwage)^2)
#mean square error
```

MSE =
2187841612

Men

Women

		Men Estimate	Men Std. Error	Men t value	Men Pr(>|t|)		Women Estimate	Women Std. Error	Women t value	Women Pr(>|t|)	
age		4.621e+02	1.139e+01	40.590	< 2e-16	***	2.162e+02	7.296e+00	29.629	< 2e-16	***
wkswork1		7.267e+02	1.616e+01	44.955	< 2e-16	***	4.377e+02	7.929e+00	55.207	< 2e-16	***
sch14	Associate	1.557e+04	2.321e+03	6.709	1.97e-11	***	1.039e+04	1.775e+03	5.851	4.89e-09	***
sch16	Bachelor	3.255e+04	2.302e+03	14.140	< 2e-16	***	1.982e+04	1.769e+03	11.204	< 2e-16	***
sch18	Advanced Degree	5.356e+04	2.314e+03	23.151	< 2e-16	***	3.310e+04	1.776e+03	18.637	< 2e-16	***

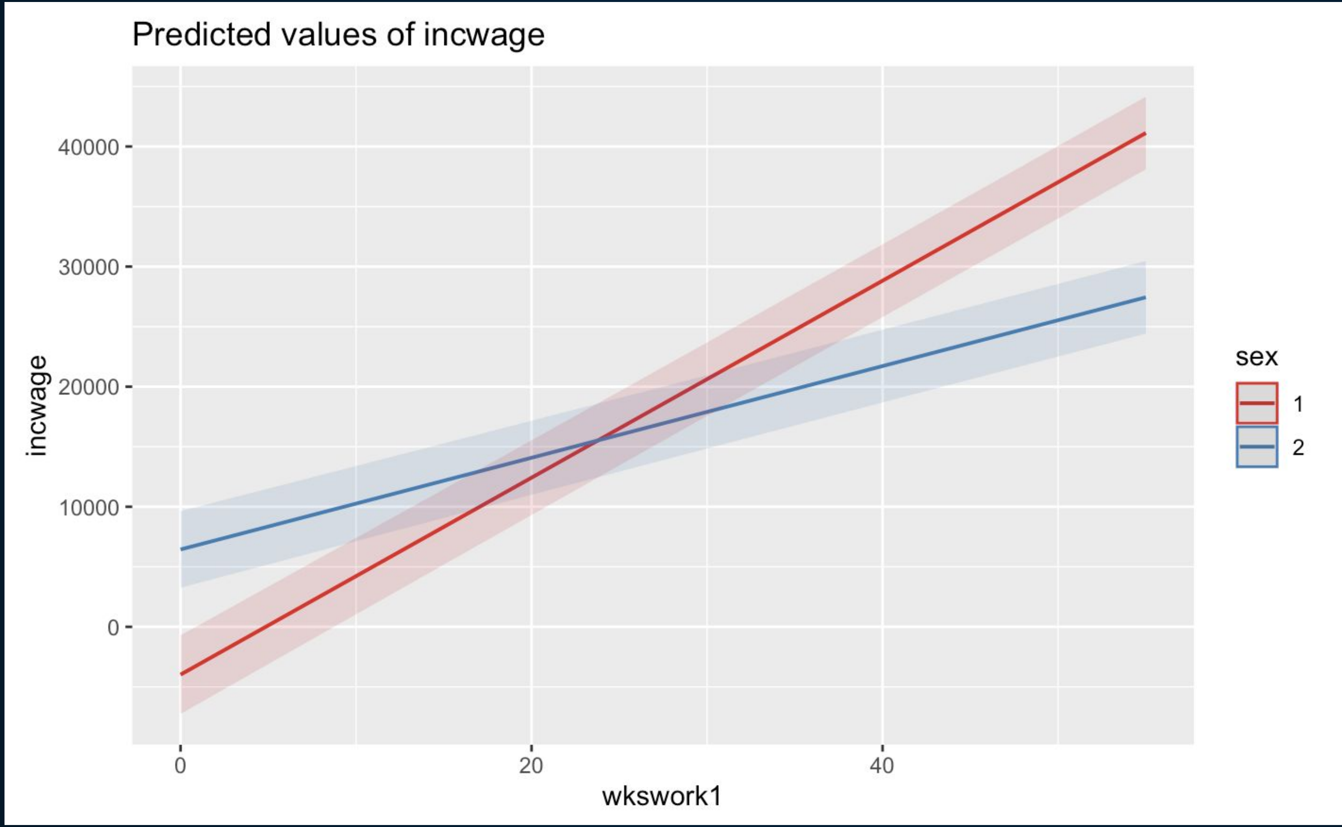
1. One Year Increase in Age: Men's Salaries Increase at a Rate Twice that of Women
2. Every Week worked in a Year: Men's Salaries Increase almost Twice as much as Women's
3. Increase in Education Level: Higher Rate of Salary Increase for Men

04

Visualization

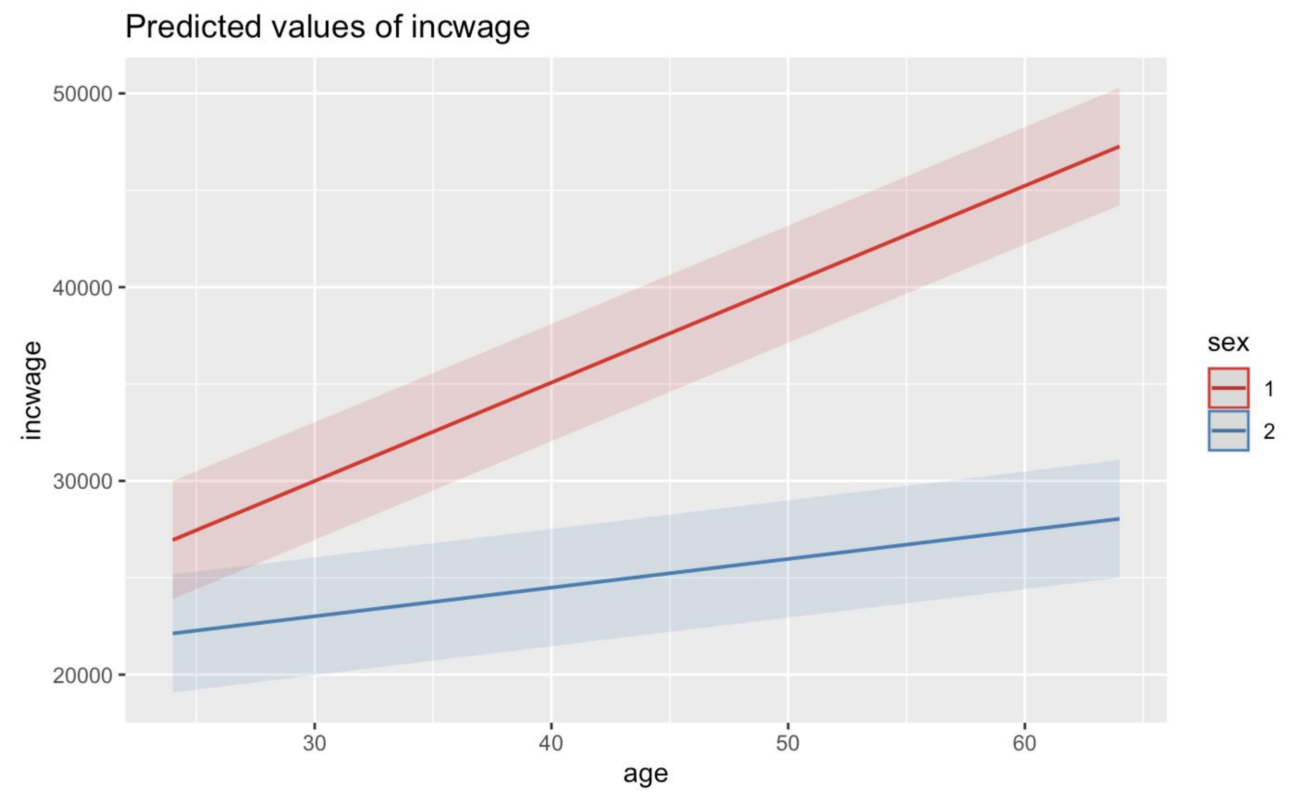
Interaction Plot

incwage~year+region+age+race+wkswork1*sex+uhrswork+sch+marst



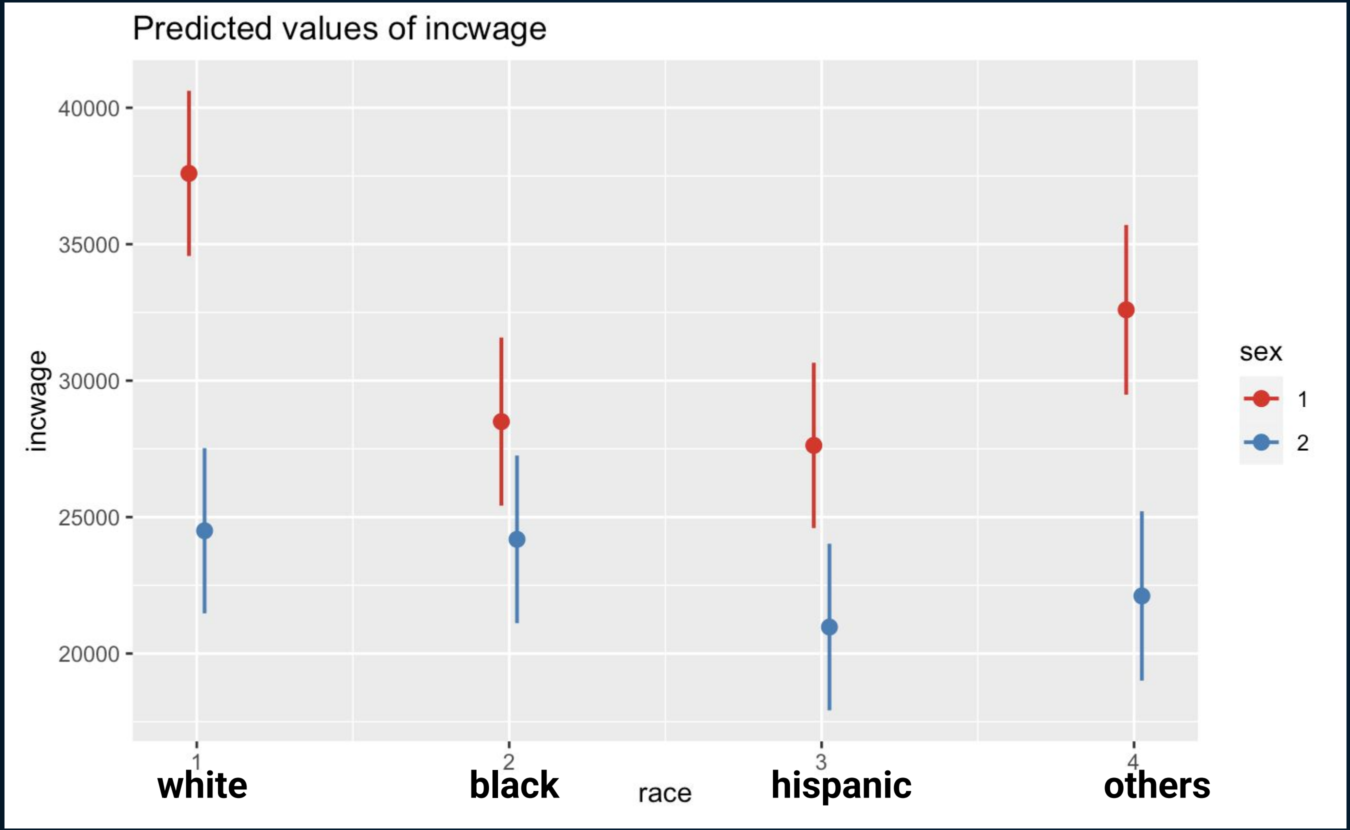
sex = 1: Male
sex = 2: Female

incwage~year+region+age+race+age*sex+uhrswork+wkswork1+sch+marst



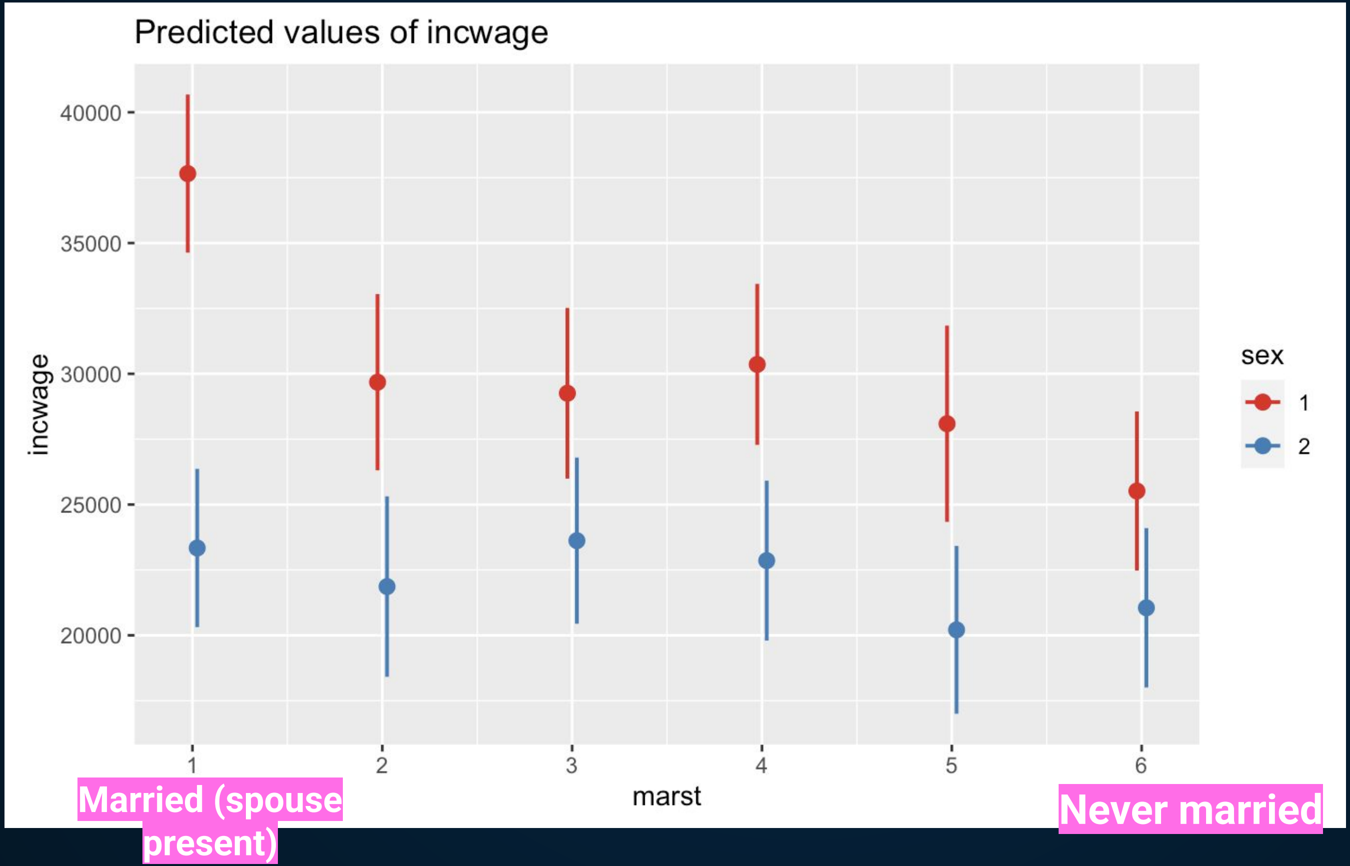
sex = 1: Male
sex = 2: Female

incwage~year+region+age+race*sex+uhrswork+wkswork1+age+sch+marst



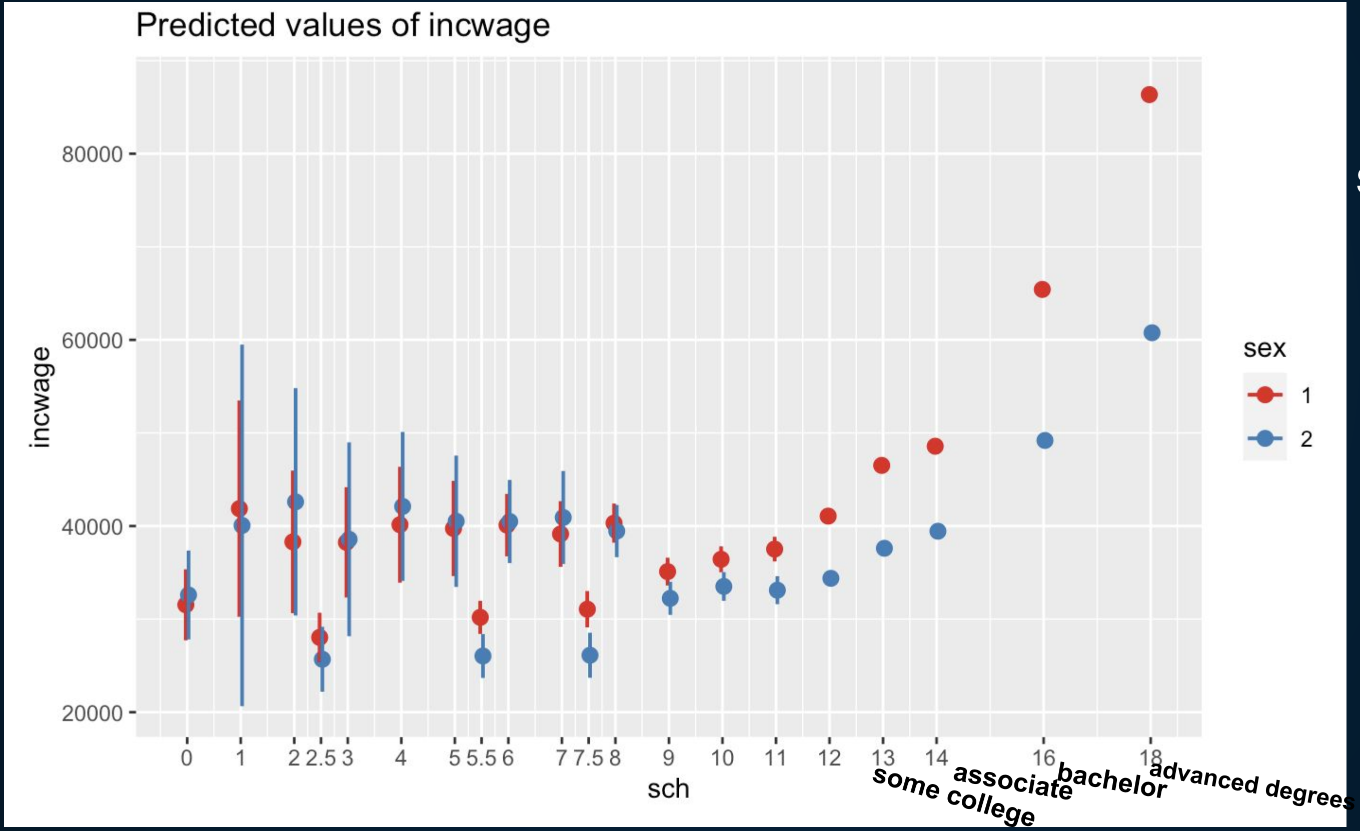
sex = 1: Male
sex = 2: Female

incwage~year+region+age+race+uhrswork+wkswork1+age+sch+marst *sex



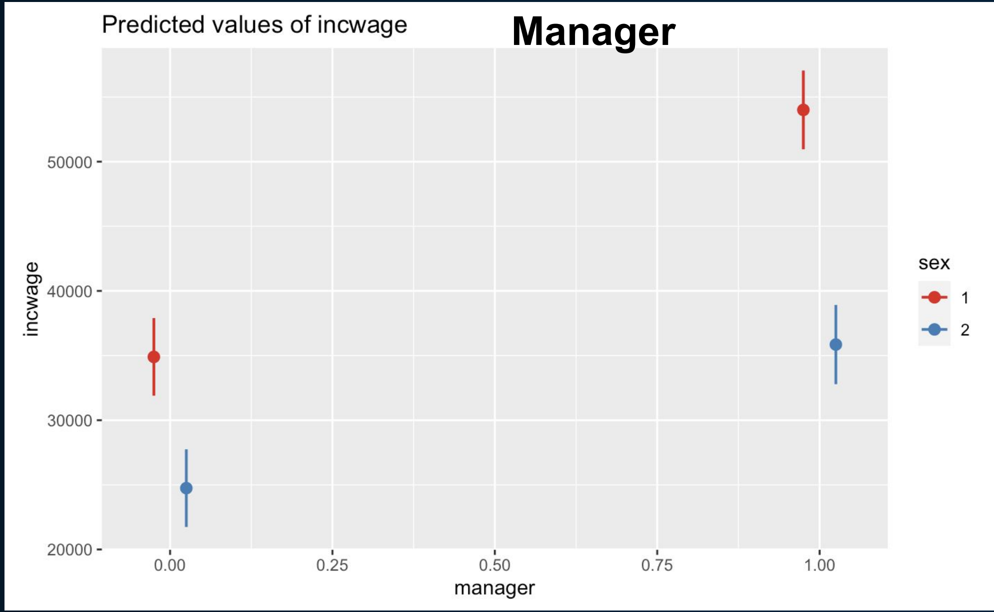
sex = 1: Male
sex = 2: Female

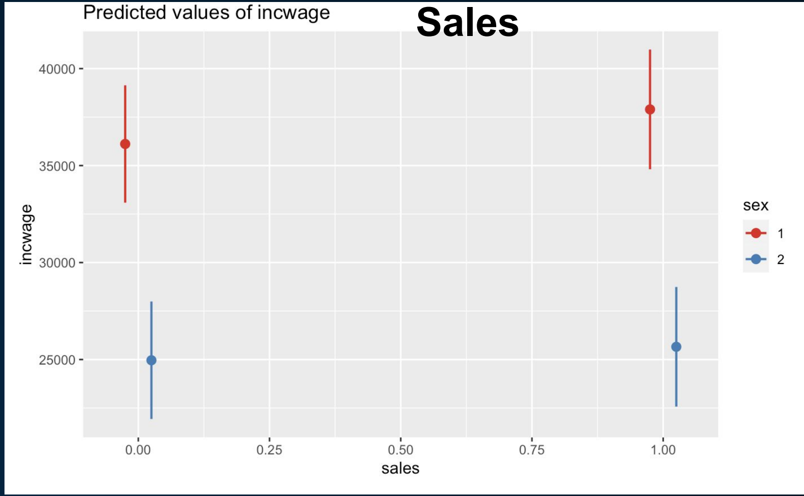
incwage~year+region+age+race+uhrswork+wkswork1+age+sch*sex+marst

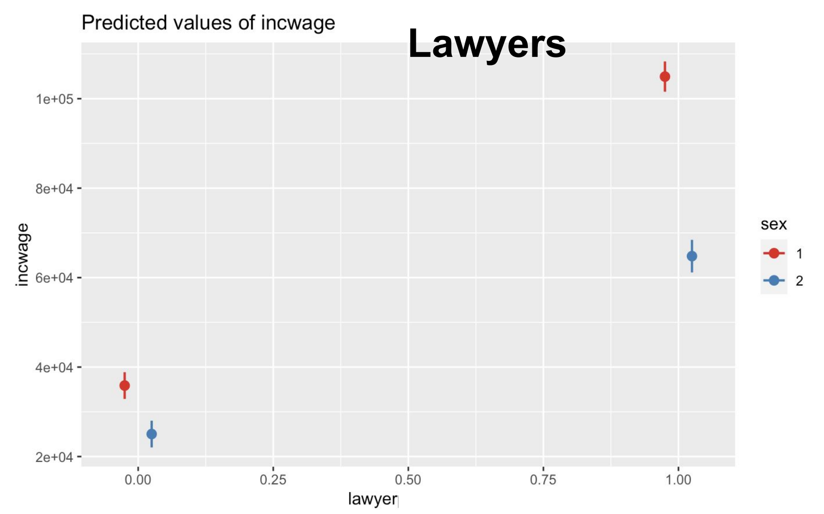


sex = 1: Male
sex = 2: Female

Industry





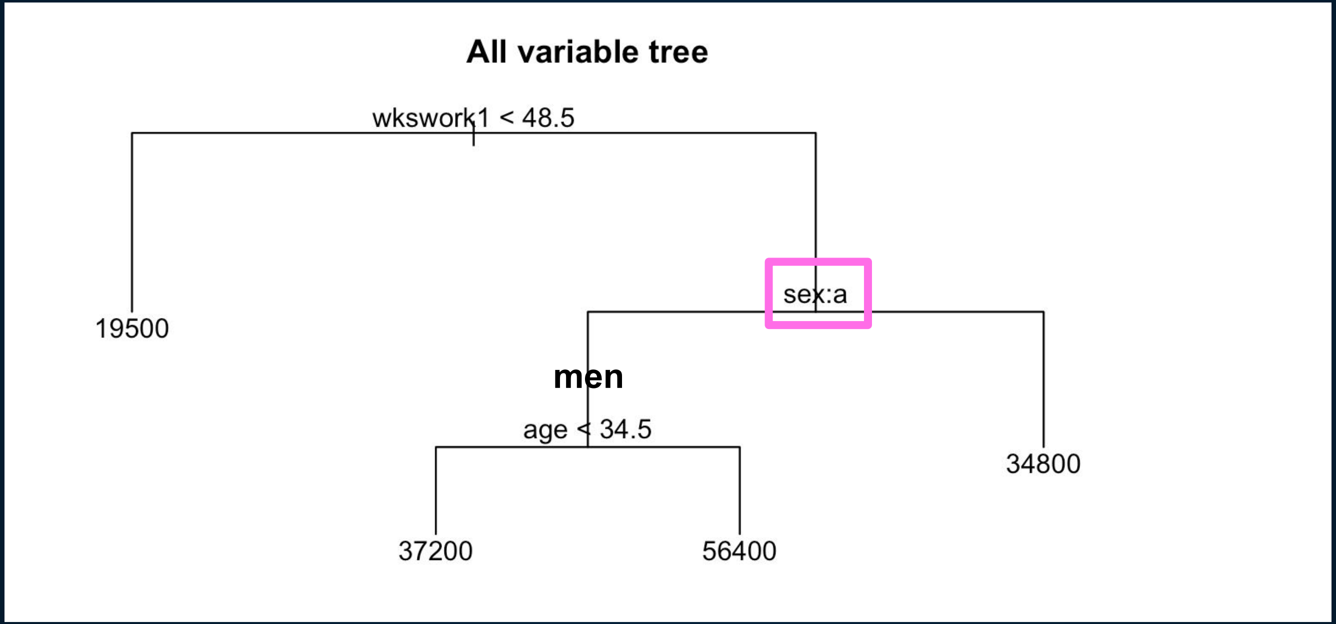


sex = 1: Male, sex = 2: Female

05

Random Forest

Single Tree



incwage~sex+age+race+wkswork1+marst+race

Random Forest

Process: Random Forest

Input

test_data

```
predicted_classes <- predict(fit.rf, newdata =
test_data)

prediction_forest <- predict(fit.rf, newdata =
test_data)
mse_forest <- mean((prediction_forest -
test_data$incwage)^2)
mse_forest
```

Output

MSE:
1870032551

06

Conclusion

Where Does the Wage Gap Exist?

Male Dominated Industries

Women are paid less in male dominated industries such as finance, law, and medicine.

Married Women

Married women are more likely to have children and are penalized for time off during and after childbirth.

Highly Educated Women

The wage gap between men and women increases as they get a higher degree of education.

Solution 1 - Reduce the wage gap between married women and men

Policies that Enable Parents to Stay in the Workforce

Offer Career Path Planning

Solution 2: Tackling Traditionally Male Dominated Industries

Promote Pay Transparency: Implement Pay Equity Audits, conduct Regular Salary Reviews

Pre-Apprenticeship Programs, Job Placements, and Ongoing Mentorship

Solution 3: Helping Women with Higher Education Level

Raising Awareness

Implement blind recruitment strategies

References

Catalyst - Workplaces that work for Women. (2023, May 31). *Women in Male-Dominated Industries and Occupations (Quick Take)*. Catalyst. https://www.catalyst.org/research/women-in-male-dominated-industries-and-occupations/

Copley, J. (2018, March 22). *Women Building Futures: Trades are a positive career option for women*. Alberta Native News. https://www.albertanativenews.com/women-building-futures-trades-are-a-positive-career-option-for-women/

Fedesoriano. (January 2022). Gender Pay Gap Dataset. Retrieved [Date Retrieved] from https://www.kaggle.com/fedesoriano/gender-pay-gap-dataset.

Frith, B. (2016, April 29). *BBC embraces 'blind' recruitment*. HR Magazine. https://www.hrmagazine.co.uk/content/news/bbc-embraces-blind-recruitment/

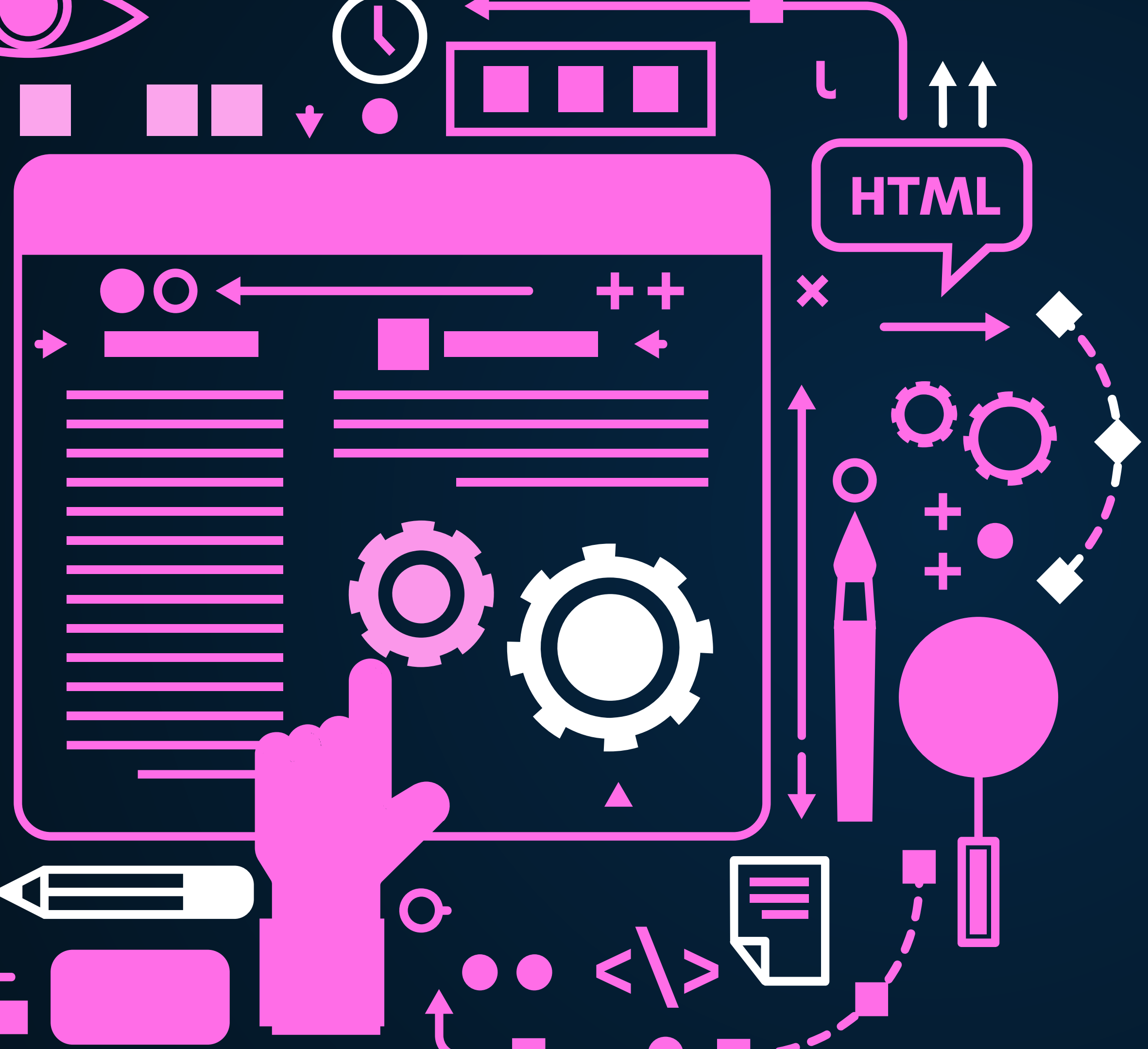
Miller, C. C. (2016, March 18). *As Women Take Over a Male-Dominated Field, the Pay Drops (Published 2016)*. The New York Times. https://www.nytimes.com/2016/03/20/upshot/as-women-take-over-a-male-dominated-field-the-pay-drops.html

Netflix Offers New Parents One Year of Paid Leave. (2015, August 5). The Wall Street Journal. https://www.wsj.com/articles/netflix-offers-new-parents-one-year-of-paid-leave-1438735806

Petriglieri, J. (n.d.). When Companies Make People Choose Between Their Career and Their Spouses'. Harvard Business Review. https://hbr.org/2018/05/talent-management-and-the-dual-career-couple

Limitations

1. US data only
2. missing values that were cleared (NAs)



THANKS !

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